all nine dimensions, rapidly and repeatedly circumnavigating the entire shape, on average making it seem as if you weren't moving through the extra six dimensions at all.

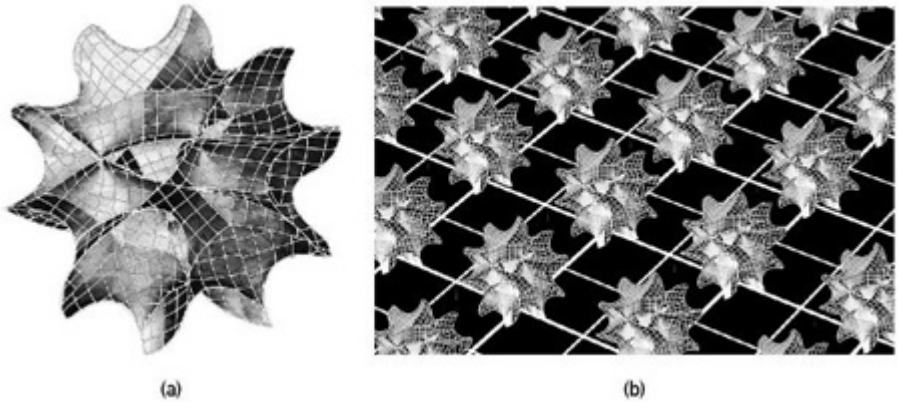


Figure 12.9: **(**a**)** One example of a Calabi-Yau shape. **(** b**)** A highly magnified portion of space with additional dimensions in the form of a tiny Calabi-Yau shape.

If these ideas are right, the ultramicroscopic fabric of the cosmos is embroidered with the richest of textures.

String Physics and Extra Dimensions

The beauty of general relativity is that the physics of gravity is controlled by the geometry of space. With the extra spatial dimensions proposed by string theory, you'd naturally guess that the power of geometry to determine physics would substantially increase. And it does. Let's first see this by taking up a question that I've so far skirted. Why does string theory require ten spacetime dimensions? This is a tough question to answer nonmathematically, but let me explain enough to illustrate how it comes down to an interplay of geometry and physics.

Imagine a string that's constrained to vibrate only on the two-dimensional surface of a flat tabletop. The string will be able to execute a variety of vibrational patterns, but only those involving motion in the left/right and back/forth directions of the table's surface. If the string is then released to vibrate in the third dimension, motion in the up/down dimension that leaves the table's surface, additional vibrational patterns